LIVEWIRE
FOR LIVE CAREERS

MATLAB for Electronics – Day wise Syllabus

DAYS	Topics 2 Hours / Day
DAY 1	**Fundamentals of MATLAB:** **Programming Script:** Introduction to working environment of MATLAB; Understanding command and workspace window – MATLAB built-in functions – Workspace variables – Array creation and concatenation – Sorting and reshaping arrays
DAY 2	Array operations – Arithmetic functions – Statistical functions – Number system conversion – Characters and strings;
DAY 3	Editor window – Decision making – If condition – Switch statement – For loop – While loop; User-defined functions;
DAY 4	2D and 3D plots – Line graphs – Bar graphs – Grouping and Stacking - Pie charts;
DAY 5	**Simulink:** Introduction to simulink interface – Library browser – Libraries and Sub Libraries; The sources library and their blocks;
DAY 6	The discontinuous library and their blocks; The sinks library and their blocks;
DAY 7	The logic and bit operations library and their blocks;
DAY 8	The lookup tables library and their blocks – Lookup table editor – Breakpoints;
DAY 9	The math operations library and their blocks;
DAY 10	The model verification library and their blocks;
DAY 11	The ports and subsystems library and their blocks;
DAY 12	The signal routing library and their blocks; Masking a simulink subsystem;

DAY 13	**Graphical User Interface (GUI):** Introduction to Graphical User Interface and their components – Building GUI based calculator – Property Inspector window – Align Objects window – Call Backs
DAY 14	Building GUI based Electronic Voting Machine (EVM)
DAY 15	**Stateflow:** Basic approach to build a Stateflow model – Junctions and Transitions – Connective junctions – Transitions priority; Control flow logic – If else condition – Switch statement – For loop – While loop – Conditions and conditions actions;
DAY 16	State actions and its working operation; Temporal logic operators and its working operation; Change detection operators and its working;
DAY 17	Chart level Functions – Types of functions – Simulink functions – MATLAB functions – Graphical functions
DAY 18	Truth tables – Conditions, decisions and actions – State transition tables – States, transitions and actions;
DAY 19	State decomposition – Exclusive OR – Parallel AND – Execution order for parallel states; Atomic subcharts – Benefits of using atomic subcharts – Map variables for atomic subcharts;
DAY 20	Masking Stateflow charts; Export chart level functions;
DAY 21	Mealy and Moore machines – Sequence detector – Vending machine; Difference between MATLAB and C as action language syntax;
DAY 22	State activity – Superstates and Substates – Child activity – Self activity;
DAY 23	**Computer Vision System Toolbox (Digital Image Processing):** **Programming Script:** Introduction to Digital Image Processing and their techniques – Image types – Image type conversion – Image arithmetic and logical operations – Image Histogram

DAY 24	Histogram equalization and Contrast stretching – Image Thresholding – Image segmentation
DAY 25	**Simulink:** RGB to Gray image conversion – Median filter – Count objects in image using labelling – Image rotation – Image complement
DAY 26	Deinterlacing – Three methods of Deinterlacing – Image sequencing and Batch processing – Object tracking – Use of correlation in Object tracking
DAY 27	Cell counting – use of Dilation and Blob analysis in Cell counting
DAY 28	**Graphical User Interface (GUI):** Image arithmetic operations – Count objects in an image
DAY 29	Image geometric transformations – Detect circular objects in an image
DAY 30	**DSP System Toolbox:** **Programming Script:** Z-Transforms – Mathematical operations on Discrete Time Signals – Even and Odd signals – Convolution – Fast Fourier Transform
DAY 31	**Simulink:** Linear Convolution – Fast Fourier Transform – Queue and Stack – Transform Time Domain Data to Frequency Domain Data – Transform Frequency Domain Data to Time Domain Data – Spectrum Analyzer
DAY 32	**Graphical User Interface (GUI):** Standard Discrete Time Signal Generation – Discrete Exponential Signal Generation
DAY 33	**Fuzzy Logic Toolbox:** **Script, Simulink & GUI:** FIS Editor – Membership Function Editor – Rule Editor – Rule Viewer - Basic tipping system – Water level controller
DAY 34	**Neural Networks Toolbox:** **Script & GUI:** Fit Data with a Neural Network – Analyze results using regression and error histogram plots – Script generation
DAY 35	Classify Patterns with a Neural Network – Analyze Pattern Recognition results using confusion matrix

DAY 36	Cluster Data with a Self Organizing Map
DAY 37	**Real Time Applications:** **Arduino Uno: Script, Simulink & GUI:** LED Blinking – Brighten and dim LED – Control Brightness of LED using POT – Control Piezo Speaker using Push Button – Dark Detector using LDR – Micro Servo Motor – Object counter using Common Anode Seven Segment Display – GUI based LDR Monitor – Modelling Laser Surgery System using Simulink
DAY 38	**Raspberry Pi: Script, Simulink & GUI:** Face Detection Algorithm – Measure and Plot temperature values using I2C Sensor – Simulink based Image Processing
DAY 39	**Internet of Things (IOT) based Arduino & Raspberry Pi Applications:** UART Serial Communication - IOT based Temperature Monitor using DHT11
DAY 40	**Interfacing with Android Sensors and Microcontrollers:** Counting Footsteps using Android Sensor – Embedded System based LED controller – HDL Coder

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